Griffith

WORLD HEADQUARTERS POUGHKEEPSIE

Memorandum to: Mr. D. W. Pendery November 5, 1956

Subject: 70 X

This memorandum presents some preliminary ideas concerning a possible new machine. Any comments, either negative or positive, will be very much appreciated.

The 70 X will be a combination 704 and 705. Ideally it will be faster than the 708 or improved 705. This would place it, in terms of speed, between the 704-705 area and the Stretch-Link area. The only design restriction on the 70 X is that it be put together out of presently available hardware, that is, cores and tubes. This dictates actual machine speeds in the same range as speeds of present machines. Any significant increase must come through machine organization.

Word Format

The 70 X will be a decimal, fixed word, machine. A word will contain 43 bits. The reasoning behind this particular word size is as follows:

1. The 70 X will have to be able to do floating point arithmetic. Ten decimal digits are approximately equal to the thirty-five bit floating point word in the 704. Ten decimal digits can be represented in forty bits. Allowing two bits to represent the signs of the characteristic and mantissa, and one bit for checking, the total bits in a floating point word are forty-three. As an additional argument for a ten decimal digit floating point system, this is the size of one of the systems used in Print 1.

2. Forty-three bits can also be used quite conveniently to represent seven six-bit alphameric characters with a check bit.

3. An instruction format can be very simple. It can be a field of ten decimal digits. One idea for a format would be to use five digits for the address, one digit for storage unit, one digit for the index register, and three digits for an operation code.

Memory

The address system allows for a memory of 100,000 words. This is probably considerably larger than is needed. The proposed 705 80,000 memory with 43 rather than 35 planes would probably be quite

sufficient considering that the 70 X will be able to store either seven or ten characters in a word as against five characters in a 705 "word." The 705 memory rather than the 704 memory is suggested because the 70 X will have a decimal system for addressing memory, very similar to the present 705 system.

Code Format

The proposed code format for six bit characters would be the same as for internal storage of six bit BCD characters in the 704. There would be only one check bit for each word. This check bit would make the total number of bits in a word odd. This would allow for an odd check on memory even when all the characters of a word are zeros.

Input and Output

Records on tape will be multiples of words as in the present 704. Tape reading and writing will be done through an ITS type unit. The 43 bits of a word will be read or written as seven characters on tape. This allows for compression and decompression of numeric fields with no effort on the part of the programmer. The printer, punch, and reader would all have plugboards which would be used to distinguish between numeric and alphameric words.

Checking

Minimum checking would consist of a check on words as they leave memory. This would be similar to the present 705 checking. Adder checking is to be hoped for if it is at all feasible. Input and output checking would be as on the present 705.

Machine Organization

The 70 X would have an adder, an M-Q, and an accumulator similar to the 704. In addition it might also have several 43 position storage units. Index registers would be similar to the 704 with the addition of some special purpose index registers which would hold one decimal digit and could be used for tape address modification.

Types of Instructions

1. Input/Output - This would be done in the manner indicated previously. Rather than having a special mark to end writing, the 70 X might have a counter (similar to MAC II in the 705) to determine the number of words to be written. Two modes of reading would also be possible. One mode would read until the end of a record on tape is

reached and add one to an index register each time. Another mode would read the number of words indicated in an index register. Time for these operations would be largely determined by tape time.

2. Arithmetic - All arithmetic operations could be done either normally or in floating point. Half word arithmetic operations would be provided for non-floating point addition and subtraction in order to use memory space more effectively. All operations will probably require the same number of machine cycles as the 704 with the possible exception of multiply and divide in either normal or floating point form.

3. Indexing - The index registers will have to be set, stored, stepped, and compared.

4. Transfers - These would be very similar to those on the 704.

5. Word Movement - Along with the usual provisions for moving information from one place in memory to another by means of placing the information in some type of storage unit, a direct memory to memory movement will be provided as in the present 705. Half word instructions will be provided for these operations. Two machine cycles will probably be required for each of these instructions as in the 704. Memory to memory instruction timing would be similar to high speed transmit in the 705.

6. M-Q and Accumulator Information Manipulation - These instructions would be similar to the 704 instructions of Left Shift, Right Shift, and Rotate M-Q.

7. Logical Operations - These instructions would be similar to the 704 instructions of And to Accumulator and Or to Accumulator. They would also include a compare instruction as in the 705.

8. Pack and Unpack Operations - These instructions would convert 4 bit characters to 6 bit characters and 6 bit characters to 4 bit characters. They would operate between memory and storage.

9. Miscellaneous Instructions - Some of the things to consider here are a Table Look Up as in the 650, Convert as in the 708, Blank Memory as in the improved 705, and Bit Manipulation as in the improved 705.

TEB:pv
cc: To Pok. Dept.

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